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# A novel markov random field segmentation method for SAR images

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**Abstract.** The synthetic aperture radar(SAR) images is difficult to be segmented by traditional Hidden Markow Random Field(HMRF)-based segmentation method, a novel HMRF-based segmentation method for SAR images is proposed in this article. The method utilizes more spacial information of SAR image model by involving the intensity differences between pixels in the potential function. An initial segmentation result is obtained by K-means method. Then accurate segmentation is carried out by the novel MRF-based segmentation method. The result showed that this method is better than the traditional HMRF one both in noise filtering and miss classification ratio.

**Keywords:** Synthetic aperture radar(SAR), image segmentation, Markov random field(MRF), EM-algorithm, potential function.

## 1. Introduction

Synthetic Aperture Radar (SAR) is an active microwave remote sensor, which is widely used in military and civilian fields due to its advantages of all-weather, high spatial resolution and strong penetrating ability. However, because the radar received signal contains a lot of noise [1], the inherent random distribution of speckle noise in the SAR image will seriously affect the quality of the image, making it very difficult to process automatically. There is currently no method that can effectively detect all features in SAR images [2,3].

Image segmentation is an important part of image analysis, but also a difficult point. Its purpose is to extract objects in images from complex backgrounds for further analysis and processing. For SAR images, image segmentation is also an important condition for target recognition or measurement. In order to overcome the influence of strong speckle noise in SAR images and achieve accurate image segmentation, the existing image segmentation methods all use a combination of coarse segmentation and refinement [4]. In the low-resolution image, the method based on threshold segmentation or clustering is used to achieve rough segmentation, and then the MRF segmentation method is used to refine the segmentation.

MRF is the most widely used statistical-based image model, and has a large number of applications in image edge extraction, segmentation, restoration, and texture analysis [5]. However, the traditional MRF method still has some defects. It does not introduce the pixel intensity value of the image

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neighborhood and the distance relationship between the pixels into the potential function, so the context information of the image is not fully utilized. [6]

In this article, the segmentation model based on MRF is used, and the EM algorithm is used to complete the image segmentation. Based on the traditional MRF segmentation model, in the neighborhood, the pixel intensity difference is introduced into the definition of the classic MRF potential function. That is, when the intensity value difference between pixels is larger, the potential function is larger, and the possibility of two pixels being classified into the same class is smaller, which provides a new idea for SAR image segmentation.

## 2. HMRF-EM segmentation model

The HMRF-EM framework was first proposed for brain MR images' segmentation [7].

If the image plane is represented as O, the data set S is$\{s_i \mid i\in O\}$. The ideal segmentation result is expressed as T=$\{t_i \mid i\in O\}$, where $t_i\in Z$, Z={1,2,…,K}, and T is a configuration of labels of S. Then the maximum posterior probability segmentation model is expressed as:

$$T^* = \underset{T}{\operatorname{argmax}}\ \{P(S|T)P(T)\}$$

P(S|T) is the likelihood probability of the relevant data set; P(T) is the prior probability of segmentation.

In the image, each pixel is related to the domain pixel, but under the condition of the known segmentation T, the pixel S is independent of each other, so we can get that

$$P(S|T) = \coprod_{I\in O} P(s_i|t_i)$$

According to the Hammersley-Clifford theorem, the Markov field T can be described by the Gibbs distribution

$$P(t_i) = \frac{1}{Q}\exp\big(-U(t_i)\big) \quad (1) \tag{1}$$

U(x) is the prior energy function.

$$U(t_i) = \sum_{d\in N} V_d(t_i) \tag{2}$$

Where N are the pixels in $t_i'$ $s$ 4-neighborhood, $V_d(t_i)$ is the clique potential, and $Q = \sum_{t_i\in T}\exp(-U(t_i))$.

If the regional data distribution is an exponential set distribution, then $P(S|T) = C_0\exp(-U(S|T,\Theta))$. Θ is the set of distribution parameters, then the maximum posterior probability segmentation model is equivalent to

$$T^* = \underset{T}{\operatorname{argmax}}\ \{U(S|T) + U(T)\}$$

Where $U(S|T) == \sum_i[\frac{(s_i-\mu_{ti})^2}{2\sigma_{t_i}^2} + \ln\sigma_{t_i}]$

Since the image data is represented as a hidden Markov process, the parameters need to be learned in an unsupervised mode, so the EM (expectation maximization) algorithm is used for modeling. The specific algorithm is as follows [8]

- Set the initial segmentation area and get the initial parameters Θ^((0))
- E-step: Calculate the conditional expectation of t+1

$$Q\big(\Theta|\Theta^{(t)}\big) = E\big\{logP(S,T|\Theta)\big|S,\Theta^{(t)}\big\}$$

- M-step: Estimate the $T^{(t+1)}$ at time t+1

$$T^{(t+1)} = \underset{T}{\operatorname{argmax}}\ \{U(S|T,\Theta^{(t)}) + U(T)\}$$

And get the distribution parameters $\Theta^{(t+1)}$ at time t+1

$$\Theta^{(t+1)} = \underset{\Theta}{\operatorname{argmin}}\, Q\left(\Theta|\Theta^{(t)}\right)$$

## 3. Proposed method

### *3.1. Image Initial Segmentation*

The segmentation performance of HMRF-EM depends on the division of the initial region, because the initial segmentation region needs to be set before the EM algorithm can be executed. Commonly used initial segmentation methods include Ostu threshold segmentation or K-means clustering.

*3.1.1. Ostu threshold segmentation.* In image segmentation, thresholding is a widely used tool. Otsu's thresholding method is an automatic and unsupervised algorithm, which is based on a discriminant measure criterion, and the optimal threshold value is obtained when the measure function is maximized. However, in the SAR image, because the noise interference is very obvious, the gray histogram distribution of the image does not necessarily appear obvious peaks and valleys. At this time, only using the threshold value obtained from the gray histogram cannot make the image segmentation get satisfactory results, and often produces serious segmentation errors.

*3.1.2. K-means clustering.* The K-means algorithm is an iterative process that aims to minimize the sum of squares J(C) of the distances from all samples in the clustering domain to the cluster centers [9]. The algorithm flow includes four steps [10], and the specific flow chart is shown in Figure 1. It can be seen that the K-means algorithm is relatively tolerant to noise.

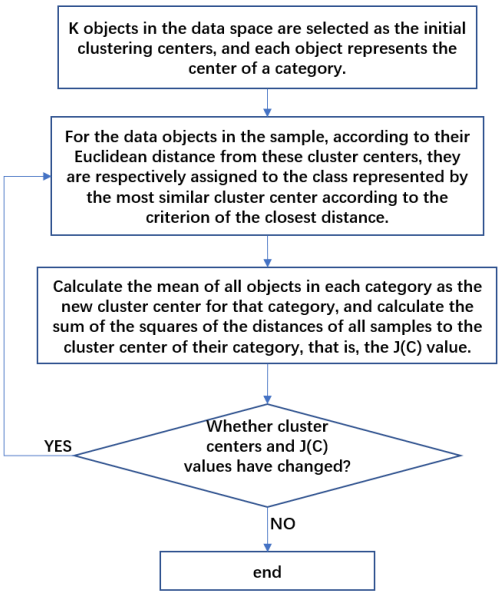



**Figure 1.** K-means algorithm flow.

Here, the SAR image is first processed by Gaussian filtering to remove part of the Gaussian noise. Then use the K-means algorithm to perform pre-segmentation to obtain the initial segmentation area and obtain the initial parameter $\Theta^{(0)}$.

Figure 2(a) is a simulated SAR image. Figure 2(b) is the simulated SAR image after Gaussian filtering, the window size is 13*13, and sigma=4. Figure 2(c) is the simulated SAR image after K-means pre-segmentation, K=2.

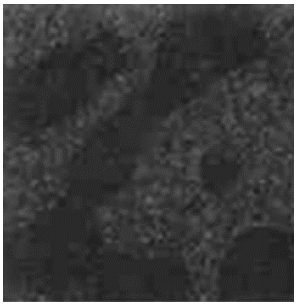


**Figure 1(a).** A simulated SAR image

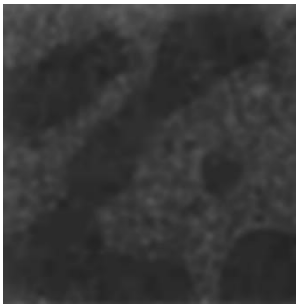


**Figure 2(b).** The simulated SAR image after Gaussian filtering



**Figure 2(c).** The simulated SAR image after K-means pre-segmentation

*3.2. HMRF-EM segmentation based on new potential energy function*

Described by the HMRF-EM algorithm, it can be seen that the context information of the image is very important for image segmentation, and the potential function is used to describe the relationship between pixels and domain pixels. Figure 3 is a graph of single-pixel and double-pixel groups in the second-order neighborhood system, according to which the potential function of the neighborhood group can be constructed.

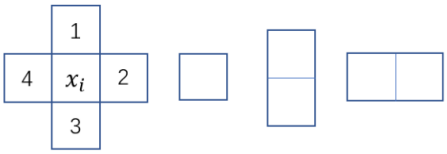



**Figure 3.** Single-pixel and double-pixel groups in the second-order neighborhood system.

However, traditional potential energy functions often cannot fully utilize contextual information. A traditional group potential function is defined as [11]

$$V_d(z_i) = \begin{cases} \beta & z_i = z_d \\ 0 & z_i \neq z_d \end{cases}$$

N is the domain pixel of $z_i$, $d \in$ N

Although this definition method simplifies the calculation, it does not consider the intensity relationship between pixels, which affects the classification effect. In view of this situation, this paper introduces a new definition formula of group potential function.

$$V_d(z_i) = \begin{cases} -\beta & z_i = z_d \\ -\frac{\sigma_i^2}{(\sigma_i^2 + \alpha(s_i - s_d)^2)}\,\beta & z_i \neq z_d \end{cases} \tag{3}$$

In the formula, $\sigma_i^2$ is the standard deviation at $z_i$, β is the classification parameter in the group, and $s_i, s_d$ are the grayscale intensities at $z_i$, $z_d$, respectively. It can be seen that when the difference between the gray intensity values at $z_i$, $z_d$ is larger, the energy between them is larger, and the possibility of being classified into one class is smaller.

According to the EM algorithm, the steps of the HMRF-EM segmentation algorithm based on the new potential energy function are as follows

Step 1 Preprocess the image through Gaussian filtering and k-means algorithm to obtain the initial segmentation area and obtain the initial parameter $\Theta^{(0)}$.

Step 2 Calculate the likelihood distribution of the nth iteration $P^{(n)}(s_i|t_i)$

Step 3 Substitute equation (3) into equation (2) to obtain the energy function U(T). Use the current parameter $\Theta^{(n)}$ and the maximum posterior segmentation to get the new current pixel category, $T^{(n)} = \underset{T}{\operatorname{argmax}} \{U(S|T) + U(T)\}$

Step 4 Find the posterior distribution of each pixel by $P^{(n)}(t_i|s_i) = \frac{P^{(n)}(s_i|t_i)P(t_i)}{P(s_i)}$, where $P(t_i)$ can be obtained from equation (1), $P(s_i) = \sum_{t \epsilon T} P(t_i)\, G(s_i; \theta_t)$, $G(s_i; \theta_t)$ is a Gaussian distribution function with parameters $\theta_t = (\mu_t, \sigma_t)$.

Step 5 Update the parameters with $P^{(n)}(t_i|s_i)$.

$$\mu_t^{(n+1)} = \frac{\sum_i P^{(n)}(t_i|s_i) s_i}{\sum_i P^{(n)}(t_i|s_i)}$$

$$(\sigma_t^{(n+1)})^2 = \frac{\sum_i P^{(n)}(t_i|s_i)(s_i - \mu_t^{(n+1)})^2}{\sum_i P^{(n)}(t_i|s_i)}$$

## 4. Experimental results

The algorithm in this paper is experimented with simulated SAR images and real SAR images, and compared with the HMRF-EM segmentation method using the traditional potential energy function. The first set of experiments is to segment the simulated SAR image into two classes, β = 20, α = 200 in the potential energy function, where (a) is the original simulated SAR image. First, it is manually segmented to obtain relatively standard segmentation results, as shown in Figure 4(b). After that, the traditional HMRF-EM segmentation method is adopted, as shown in Figure 4(c), and segmented by the method proposed in this paper, as shown in Figure 4(d). It can be seen from the experimental results that the segmentation effect of this method is obviously better than that of the traditional MRF algorithm.

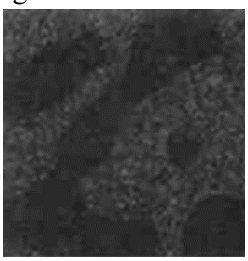


**Figure 4(a).** The original simulated SAR image.



**Figure 4(b).** The standard segmentation results.

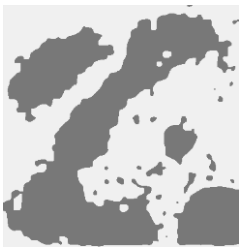


**Figure 4(c).** Using the traditional HMRF-EM segmentation method.



**Figure 4(d).** Using the HMRF-EM segmentation method in this picture.

The second set of experiments is to segment real SAR images with β = 20, α = 200 in the potential energy function, where (a) is the original simulated SAR image. First, it is manually segmented to obtain relatively standard segmentation results, as shown in Figure 4(b). After that, the traditional HMRF-EM segmentation method is adopted, as shown in Figure 4(c), and segmented by the method proposed in this paper, as shown in Figure 4(d). As can be seen from Figure 4(c), the traditional HMRF-EM segmentation method cannot deal with the noise in the picture well, resulting in more misclassified pixels. However, the method in this paper has fewer segmentation errors due to noise, and the anti-noise effect is far superior to the traditional HMRF-EM segmentation method.



**Figure 5(a).** The original simulated SAR image.



**Figure 5(b).** The standard segmentation results.



**Figure 5(c).** Using the traditional HMRF-EM segmentation method.



**Figure 5(d).** Using the HMRF-EM segmentation method in this picture.

## 5. Conclusion

For the segmentation of SAR images, based on the traditional MRF segmentation method, this paper introduces the pixel grayscale difference information into the potential function of the image neighborhood. That is, the greater the grayscale difference of the pixels, the less likely they are to be classified into the same class. In this way, the spatial information of the image is more fully utilized. The HMRF-EM segmentation model is used to iteratively obtain classification predictions under unsupervised conditions. For the problem that there is a lot of noise in the SAR image, which is difficult to deal with, the K-means algorithm is selected for pre-segmentation to obtain the initial area division. It avoids the serious misclassification problem caused by the fact that the grayscale histogram distribution of the image has no obvious peaks and valleys because of the noise, when we use Otsu threshold segmentation. The proposed method and the traditional HMRF-EM segmentation method are compared in experiments. It can be seen that the SAR image segmentation method proposed in this paper has obvious improvement in noise suppression and segmentation accuracy compared with the traditional HMRF-EM method.

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